

Sean Lange

Full-Stack Software Engineer

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CERTIFICATIONS & EDUCATION

University of South Florida
Bachelor of Science, Computer Science

Tampa, FL
Expected May 2028

IBM Certified Professional Developer – Cloud v6
Cloud Infrastructure, Deployment Pipelines, Distributed Systems

Orlando, FL
October 2025

PROJECTS

Multyx.io  – Multiplayer Game Development Platform – Founder

February 2025 - Present

Founded a browser-based multiplayer game development platform enabling beginner programmers to build, test, and deploy real-time games without networking knowledge, running a closed-beta with 8 testers.

Designed a **block-based visual programming model** to eliminate shared mutable state and preventing race conditions.

Implemented ECS on EC2 instead of ECS Fargate, allowing a **43% reduction in per-instance hosting costs**, by enabling high-density, dynamically routed multiplayer servers with granular control over compute resources used by each instance.

Architected stateless game container networking using ALB dynamic port routing, ECS task registration, and a custom reverse proxy, enabling **horizontal scaling that rapidly connects users** through a Redis cache storing game IP and port.

Built a Step Functions-orchestrated provisioning pipeline to reliably build, deploy, and terminate per-game server instances, achieving **~40 seconds** between request time to live server.

Designed a **typed AST model** to enable editor-time logic validation and prevent invalid logic before testing or deployment.

Implemented container-level monitoring to automatically shut unhealthy instances and allow a usage-based pricing model.

Fitted  – Outfit Generation Webapp – Creator

April 2025 – April 2025

Led a team of 4 to win AI track (**1 of 50**) at HackUSF; shipped full-stack web app in **24 hours**.

Built an AI outfit generation webapp using TypeScript and Next.js, with a type-safe tRPC API layer and PostgreSQL storage.

Designed a clothing embedding model to formalize style, appearance, and functionality, utilizing a cosine-similarity search layer over embedded clothing vectors, enabling a **sub-200ms semantic outfit matching** system across wardrobe images.

Deployed **scalable asset storage and delivery** with AWS S3 and CloudFront caching to eliminate clothing item load times.

Imgscrpr  – Adaptive Reddit Feed for Discord – Creator

October 2021 - February 2022

Built and monetized a Discord bot added by **100+ servers**, converting several users to a paid tier.

Implemented **rate-limit aware batching and caching strategy** to remain compliant with Reddit API policies while supporting 100+ concurrent servers, along with providing a **<2 second response time** for requests from the Discord API.

Designed tiered feature gating and Patreon billing pipeline to convert free servers into a **recurring-revenue** pricing model.

Scaled to **maximum server-cap** for unverified bots in one week, shut down after hitting third-party data policy constraint.

SKILLS

Languages & Concepts

JavaScript, TypeScript, Python, JSON, YAML, HTML, CSS, Git, WebSockets, REST APIs, Bash

Frameworks

React.js, Express.js, Next.js, tRPC, Auth.js, Tailwind CSS

Infrastructure

AWS (ALB, EC2, ECS, IAM, Lambda, Step Functions, VPC), Docker, CI/CD, Playwright

Data

PostgreSQL, Redis, MongoDB, AWS (S3, ElastiCache, CloudFront), Drizzle, Prisma

WORK EXPERIENCE

Publix Supermarkets

Largo, FL / Tampa, FL

Customer Service Staff

March 2022 – Present

Trusted with operating high-value money transactions (**\$3000+**) and managing the end-of-day reconciliation for the store.

Coordinated teams of 10+ people, ensuring high efficiency through varying traffic and peak \$20,000/hour revenue periods.